



Computer Networks

Ch.3: Understanding Subnetting

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Ch.3: Outline

- ☐ Introduction
- ☐ IP Addressing Overview.
- ☐ Fundamentals of Subnetting.
- ☐ Calculating Subnets, ranges, and masks.
- ☐ Apply subnetting to real-world scenarios.
- ☐ Advanced Subnetting Techniques.

Introduction

➤ What is Subnetting?

- Subnetting is the process of breaking down a larger network into smaller subnetworks (subnets), enhancing organization, performance, and scalability.

➤ Purpose:

- Efficient IP address utilization.
- Minimized network traffic.
- Enhanced security and isolation.

Introduction

➤ Why Subnetting Matters

- Essential for IPv4, where address space is limited.
- Integral in modern network design for scalability and control.

➤ Real-World Scenario:

- A large office needs isolated networks for HR, Sales, and IT. Subnetting separates these logically within the same IP address block.

➤ Impact: Better resource management, fewer bottlenecks, and scalability.

IP Addressing Overview

➤ IPv4 Address Structure

- Composed of 32 bits:

- Network part: Identifies the network.
- Host part: Identifies individual devices.

- Example:

192.168.1.1 → 11000000.10101000.00000001.00000001

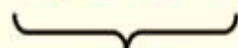
- Classes:

- Class A: /8, large networks (+16M hosts).
- Class B: /16, medium networks (+65K hosts).
- Class C: /24, small networks (256 hosts).

IP Addressing Overview

IP Address

192.168.1 .1



Octet $\equiv$ 8bit

(0 , 1, 2, 3, ..., 255)

Length $\equiv$ 8 X 4 $\equiv$ 32 bit

1-Network Address



2-Host Address



Private vs. Public IPs

➤ Private IPs

- Not routable on the internet. Used within organizations.
- First octet: Determine the class of IP Address.
- Classes of IP Addresses:
 - Class A: **1.0.0.0** to **126.0.0.0** (Large networks).
 - Class B: **128.0.0.0** to **191.255.0.0** (Medium networks).
 - Class C: **192.0.0.0** to **223.255.255.0** (Small networks).

➤ Public IPs

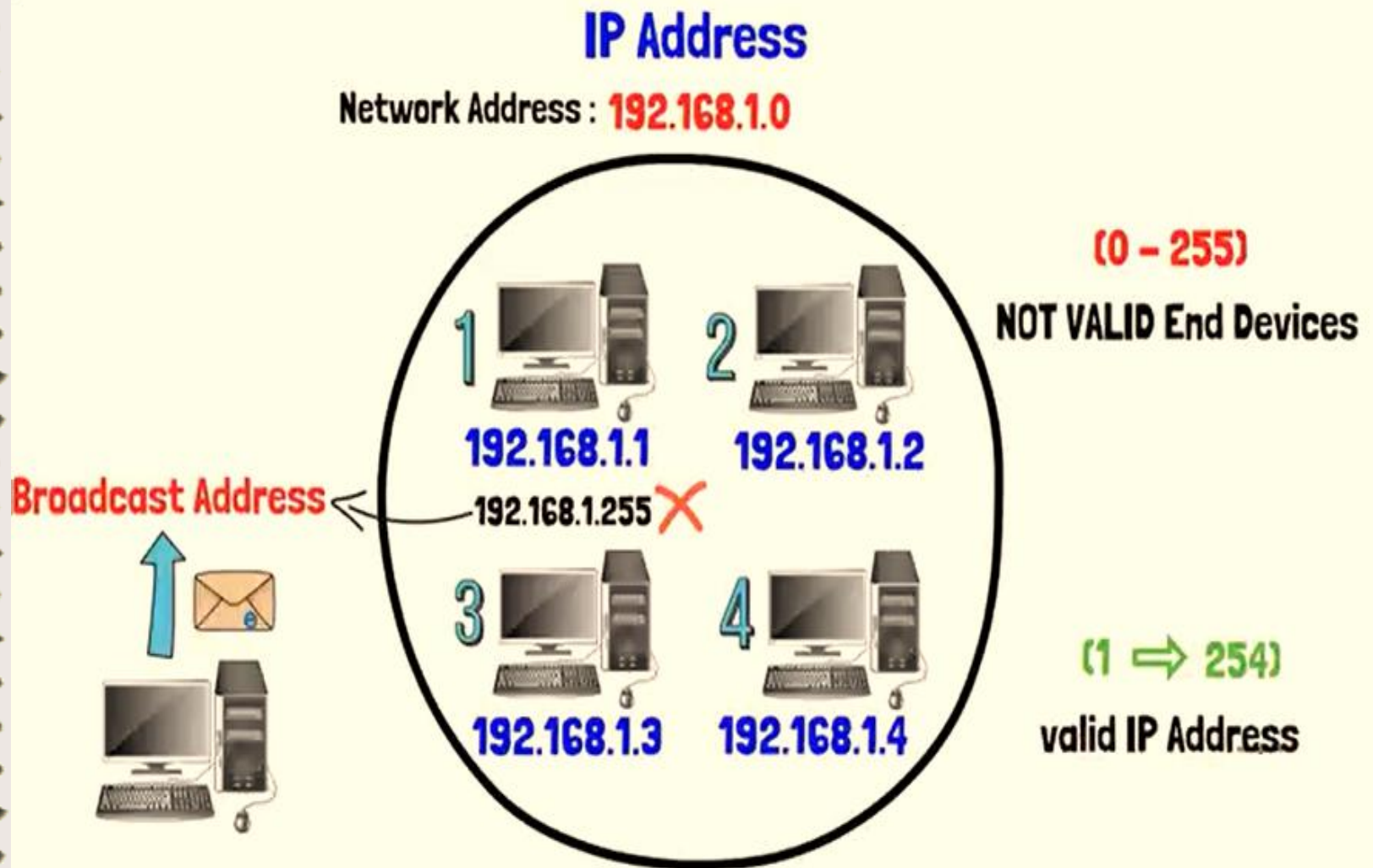
- Globally unique and routable on the internet.

Private IP Address Classes

Private IP Address Classes

Class	IP Address Range	Default Subnet Mask	For
Class A	10.0.0.0 - 10.255.255.255	^N ^H ^H ^H 255.0.0.0	Large Organisation
Class B	172.16.0.0 - 172.31.255.255	^N ^N ^H ^H 255.255.0.0	Mid size Organisation
Class C	192.168.0.0 - 192.168.255.255	^N ^N ^N ^H 255.255.255.0	home, office and small company

The Benefits of Subnetting



Subnet Masks and CIDR Notation

➤ What is a Subnet Mask?

- Defines the boundary between the network and host portions of an IP address.
- Splits an IP address into:
 - Network portion: Identifies the subnet.
 - Host portion: Identifies devices.
- Example:
 - Address: 192.168.1.1
 - Subnet Mask: 255.255.255.0
 - Binary: 11111111.11111111.11111111.00000000

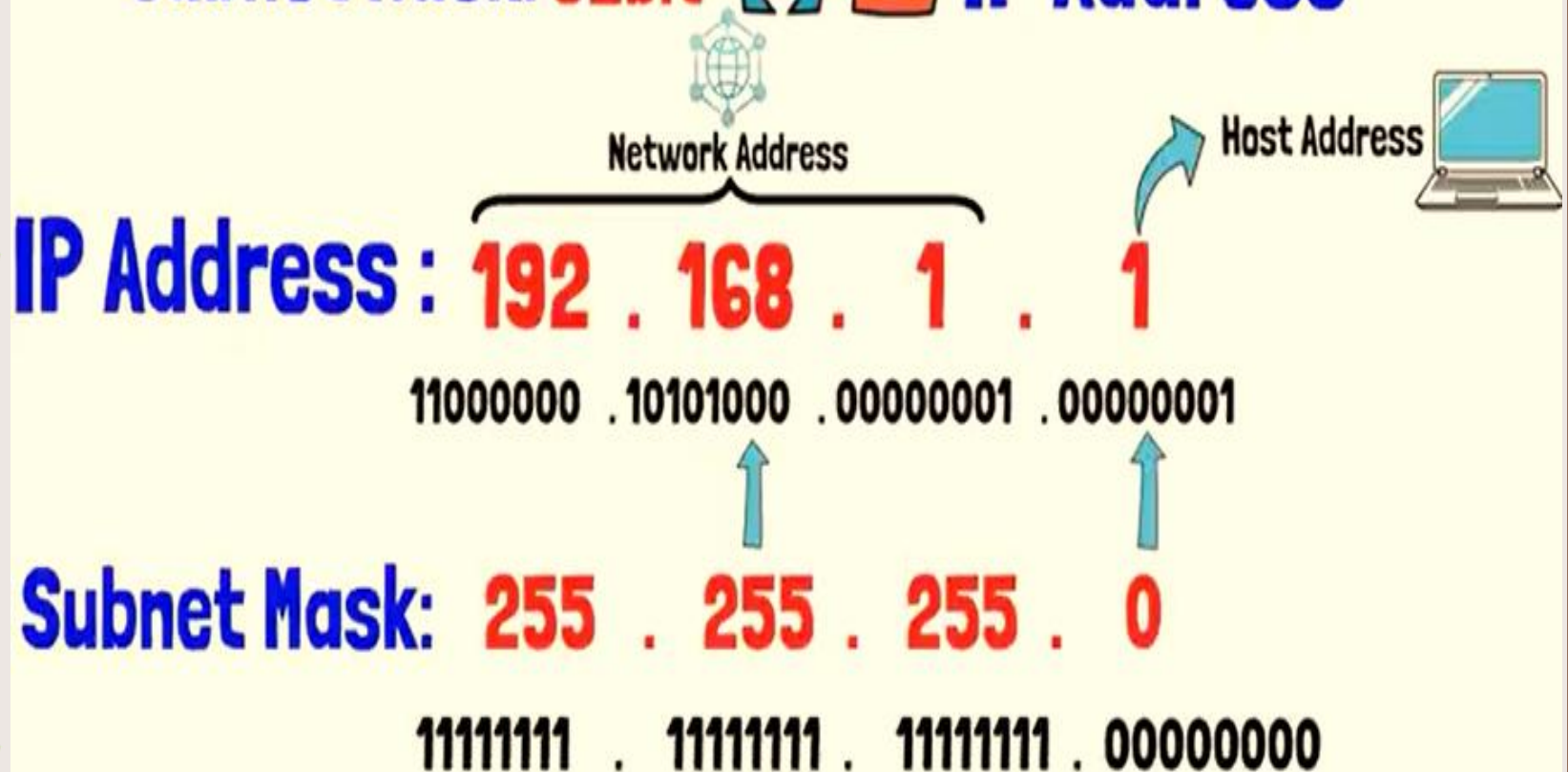
Subnet Masks and CIDR Notation

➤ CIDR (Classless Inter-Domain Routing) Notation

- A shorthand for subnet masks.
- Compact form of subnet mask representation.
- Example:
 - Subnet Mask: 255.255.255.0
 - CIDR: /24 (24 bits for network, 8 bits for hosts).
 - Again: /24 = 24 bits for the network, leaving 8 bits for hosts.

Subnet Masks

Subnet Mask: 32bit  IP Address



The Benefits of Subnetting

- Efficient Use of IP Addresses:
 - Avoids wastage in large networks.
- Improved Performance:
 - Smaller broadcast domains reduce congestion.
- Security and Isolation:
 - Restrict communication between subnets.
- Simplifies Network Management:
 - Logical segmentation of departments or services.

How Subnetting Works

1) Borrowing Bits:

- Take bits from the host portion for additional subnets.

2) Subnet Calculation:

- $2^n \rightarrow$ Number of subnets (n = bits borrowed).
- $2^h - 2 \rightarrow$ Number of hosts (h = remaining host bits).

3) Assign Ranges:

- Define network, broadcast, and usable IP addresses.

Subnetting Formula

- **Total Subnets** = 2^n , where n is borrowed bits.
- **Total Hosts** = $2^h - 2$, where h is host bits.
- **Total IPs in Subnet:** 2^h .
- **Example:**
 - Network: 192.168.1.0/24
 - Borrow 2 bits → /26.
 - Subnets: $2^2 = 4$.
 - Hosts per subnet: $2^6 - 2 = 62$

Example Scenario 1

- Ex1: Suppose the Network 192.168.1.0/24.
Divide the given network into 4 subnets.

Step-by-Step Solution:

- Determine Bits to Borrow:
 - $2^n \geq 4$, so $n = 2$.
 - New subnet mask: /**26** (255.255.255.192).
- Calculate Subnet Size:
 - Host bits left: 6 ($2^6 =$ **64**).
 - Each subnet has 64 addresses (1 network, 1 broadcast, **62** usable).

Example Scenario 1 (Cont.)

- Subnets table:

Subnet	Network ID	Start IP Address	End IP Address	Broadcast
1	192.168.1.0	192.168.1.1	192.168.1.62	192.168.1.63
2	192.168.1.64	192.168.1.65	192.168.1.126	192.168.1.127
3	192.168.1.128	192.168.1.129	192.168.1.190	192.168.1.191
4	192.168.1.192	192.168.1.193	192.168.1.254	192.168.1.255

Example Scenario 2

- Ex2: Suppose the Network 172.16.0.0/16.
Subnets with at least 200 hosts.

Step-by-Step Solution:

- Determine Bits for Hosts:
 - $2^h - 2 \geq 200 \rightarrow h = 8.$
 - Remaining bits for subnetting: $32 - 16 - 8 = 8.$
 - New subnet mask: /24 (255.255.255.0).
- Calculate Subnet Count:
 - Total subnets: $2^8 = 256.$

Example Scenario 2 (Cont.)

➤ Subnets table:

Subnet	Network ID	Start IP Address	End IP Address	Broadcast
1	172.16.0.0	172.16.0.1	172.16.0.254	172.16.0.255
2	172.16.1.0	172.16.1.1	172.16.1.254	172.16.1.255
3	172.16.2.0	172.16.2.1	172.16.2.254	172.16.2.255
...	...	...	...	...
256	172.16.255.0	172.16.255.1	172.16.255.254	172.16.255.255

Example Scenario 3

➤ Ex3: Suppose the Network 192.168.10.0/24.
Divide the network into 2 Departments:

- Department A: 120 hosts.
- Department B: 50 hosts.

Step-by-Step Solution:

- Department A:
 - Hosts: $2^h - 2 \geq 120 \rightarrow h = 7$.
 - Subnet mask: /25 (255.255.255.128).
 - Subnet: 192.168.10.0 - 192.168.10.127
 - Usable IPs: 192.168.10.1 - 192.168.10.126.

Example Scenario 3 (Cont.)

■ Department B:

- Hosts: $2^h - 2 \geq 50 \rightarrow h = 6$.
- Subnet mask: /26 (255.255.255.192).
- Subnet: 192.168.10.128 - 192.168.10.191
- Usable IPs: 192.168.10.129 - 192.168.10.190.

Subnet	Network ID	Start IP Address	End IP Address	Broadcast
A	192.168.10.0	192.168.10.1	192.168.10.126	192.168.10.127
B	192.168.10.128	192.168.10.129	192.168.10.190	192.168.10.191

Real-World Applications

➤ Enterprise Use Case:

- Segmenting departments for better management.
 - Segmenting HR, IT, and Sales.

➤ ISPs:

- Allocate IP blocks to customers.

➤ Cloud Networking:

- Subnets in a Virtual Private Cloud (VPCs).
 - AWS or Azure.

Advanced Subnetting Techniques

- Variable Length Subnet Masking (VLSM):
 - Allows subnets of different sizes.
- Supernetting:
 - Combines smaller networks into larger ones (CIDR).
- IPv6 Subnetting:
 - Supports massive address space (128 bits).

Conclusion

- Practice binary math for better accuracy.
- Subnetting enhances network efficiency, performance, and security.
- A systematic approach ensures accurate subnet design.
- Mastery comes with practice and real-world application.

A silver metal spiral binding is visible along the left edge of the page, with the wire looping through a series of holes.

Any Questions?